

Lipid-soluble Extracts from *Salvia miltiorrhiza* Inhibit Production of LPS-induced Inflammatory Mediators via NF- κ B Modulation in RAW 264.7 Cells and Perform Antiinflammatory Effects *In Vivo*

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Salvia miltiorrhiza, a traditional Chinese herbal medicine, is used to treat various inflammatory diseases. In the present study, the antiinflammatory effects of *S. miltiorrhiza* lipid-soluble extracts (SMLE) were demonstrated *in vitro* and *in vivo*, along with its underlying mechanism of action. SMLE significantly inhibited the production of NO, TNF- α , IL-1 β and IL-6 in lipopolysaccharides (LPS)-stimulated RAW 264.7 cells. SMLE also inhibited the LPS-induced degradation of I κ B- α in the cytoplasm and the translocation of p65 to the nucleus in RAW 264.7 cells. In addition, SMLE inhibited the production of intracellular reactive oxygen species (ROS) and the surface expression of CD14 induced by LPS. In animal models, intraperitoneal administration of SMLE increased the survival rate of endotoxemia and sepsis in mice. The topical administration of SMLE significantly inhibited ear edema induced by PMA. It was found that SMLE inhibits the LPS-induced gene and protein expression of iNOS, TNF- α , IL-1 β and IL-6 in macrophages by blocking NF- κ B activation, and these effects are mediated, at least in part, through the inhibition of intracellular ROS generation and the surface expression of CD14. The results suggest a possible therapeutic application of SMLE in inflammatory diseases and provide scientific evidence in support of the traditional Chinese medical practice of treatment with *S. miltiorrhiza*. Copyright © 2012 John Wiley & Sons, Ltd.

Keywords: *Salvia miltiorrhiza*; antiinflammatory effect; RAW 264.7 cells; inflammatory mediators; NF- κ B.

INTRODUCTION

Macrophages play an important role in inflammatory disease through the release of cytokines such as tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), interleukin-6 (IL-6), nitric oxide (NO) and other inflammatory mediators (Adams and Hamilton, 1984). The production of these mediators has been observed in many tissues following exposure to immune stimulants, including the bacterial endotoxin lipopolysaccharides (LPS). In response to inflammatory stimuli such as reactive oxygen species (ROS), LPS and cytokines, I κ B- α is phosphorylated and released from nuclear factor- κ B (NF- κ B). Activated NF- κ B p65 migrates to the nucleus and up-regulates inflammation-related genes such as iNOS and pro-inflammatory cytokines (Gius *et al.*, 1999; Allen and Tresini, 2000). Excessive production of inflammatory mediators is involved in systemic inflammation, such as endotoxemia and sepsis, as

well as local inflammation, such as skin and soft tissue infections. Inhibition of the production of inflammatory mediators in response to inflammatory stimuli might be a useful therapeutic strategy in treating inflammatory diseases (Nam, 2006; Wang *et al.*, 2007).

The rhizome of the herb *Salvia miltiorrhiza* Bunge, also known as Danshen, is known for promoting the circulation of blood and eliminating stasis in traditional Chinese medicine. *S. miltiorrhiza* extracts contain lipid-soluble diterpene quinones (tanshinones) and water-soluble phenolic acid derivatives, such as tanshinol and salvianic acid (Shi *et al.*, 2005). Recent articles have shown that lipid-soluble extracts of *S. miltiorrhiza* had preventive effects on hepatocyte injury (Park *et al.*, 2009), antiatherogenic properties (Zhang *et al.*, 2008), inhibitory effects on the proliferation of breast cancer cells (Yang *et al.*, 2010) and anti-relapse properties in alcohol-preferring rats (Serra *et al.*, 2003). The oil and liquor agents of *S. miltiorrhiza* have been used to treat local and systemic inflammation for over 1500 years (Zhonghua Bencao Editors, 1998). Although the *in vitro* antiinflammatory effects of some pure compounds from *S. miltiorrhiza* have been reported (Jang *et al.*, 2006; Tang *et al.*, 2011; Jeon *et al.*, 2008), the lipid-soluble extract is poorly understood. It was previously demonstrated that the water-soluble extract

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contributed less than the lipid-soluble extract to the reduction of NO production in LPS-induced RAW 264.7 cells (unpublished results from the author Run-Lan Cai). The present study explored the effects of the *S. miltiorrhiza* lipid-soluble extract (SMLE) on inflammation *in vivo* and on the production of inflammatory mediators in RAW 264.7 cells stimulated with LPS. Furthermore, the underlying molecular mechanisms of these effects were investigated.

MATERIALS AND METHODS

Reagents. Tanshinone IIA, cryptotanshinone, tanshinone I and 15,16-dihydrotanshinone I (purity $\geq 98\%$) were purchased from the National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). iNOS, NF- κ B p65, I κ B- α and β -actin antibodies were obtained from Santa Cruz Biotechnology, Inc. (Santa Cruz, CA, USA). Oligonucleotide primers and dNTP mix were purchased from Bio Basic Inc. (Bio Basic, Toronto, Canada). PE-rat IgG_{2a}, PE-rat anti-mouse TLR4/MD complex, mouse BD Fc block, FITC rat IgG₁ and FITC rat anti-mouse CD14 were obtained from BD Pharmingen (BD Pharmingen, San Diego, CA). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT), LPS, phorbol myristate acetate (PMA), 2, 7-dichlorofluorescein diacetate (DCFH₂-DA) and all other chemicals were purchased from Sigma Chemical Co. (St Louis, MO, USA).

Animals. Balb/c mice (male, 18–20 g) and ICR mice (male, 20–22 g) were purchased from Vital River Experimental Animal Services (Beijing, China). Mice were kept in standard laboratory conditions of temperature and humidity with a 12 h light/dark cycle. All experiments were carried out according to the National Institutes of Health Guide for the Care and Use of Laboratory Animals and were approved by the Animal Ethics Committee of the Institute of Medicinal Plant Development, Chinese Academy of Medical Sciences.

Preparation of *S. miltiorrhiza* lipid-soluble extracts. The herb used in this study was identified as the rhizome of *S. miltiorrhiza* by Professor Xian'en Li of the Institute of Medicinal Plant Development (IMPLAD), Chinese Academy of Medical Sciences. A voucher specimen was deposited in the Herbarium of IMPLAD. Air-dried whole rhizome of *S. miltiorrhiza* (500 g) was ground to a powder and extracted by refluxing with an ethanol–water (95:5, v/v, 5 L) solution for 2 h, and then repeated two times. The supernatant was concentrated to remove the ethanol solvent under vacuum, and the residue was precipitated with ethanol–water (15:85, v/v). After placing the sample at 4 °C for 48 h, the precipitate was collected and vacuum dried. The final sample represented a 2.4% yield of raw material dry weight. The final SMLE was stored at 4 °C until further use.

HPLC analysis of SMLE. A Shimadzu SC-15C HPLC system equipped with LC solution software and a UV spectrophotometer detector was used for analysis. A reverse phase Intersil C₁₈ column (4.6 mm $\times$ 250 mm, 5 μ m) was used for chromatographic separation and was kept at room temperature. The mobile phase

consisted of acetonitrile–water (56.8:43.2, v/v) at a flow rate of 1.5 mL/min. The samples were dissolved in acetonitrile, and 12 μ L was injected. The UV spectrophotometer detector was set at 254 nm. Retention times for 15,16-dihydrotanshinone I, cryptotanshinone, tanshinone I and tanshinone IIA were 12.36, 19.98, 22.95 and 40.43 min, respectively.

Cell culture and sample treatment. The murine macrophage RAW 264.7 cell line was obtained from American Type Culture Collection (ATCC, Rockville, MD, USA). The cells were cultured at 37 °C in DMEM (Invitrogen, California, USA) supplemented with 10% heat-inactivated FBS (Hyclone, Utah, Logan, USA), 2 mM L-glutamine, penicillin G (100 IU/mL) and streptomycin (100 μ g/mL) in a humidified atmosphere containing 5% CO₂ and 95% air. The cells were incubated with SMLE at different concentrations and then stimulated with LPS (10 ng/mL) for the indicated time. The stock solutions of SMLE were prepared in dimethyl sulfoxide (DMSO), and the final concentration of DMSO was less than 0.5%.

NO production and cell viability measurements. The level of NO production was monitored by measuring the nitrite level in the culture medium. This was performed by mixing the medium with Griess reagent (1% sulfanilamide in 5% phosphoric acid and 0.1% N-1-naphthylethylenediamine dihydrochloride in water). The absorbance was measured at a wavelength of 540 nm after incubation for 10 min. The nitrite concentration was calculated with reference to a standard curve of sodium nitrite generated from known concentrations. Cell viability was assessed using an MTT assay.

Measurement of TNF- α , IL-1 β and IL-6 in the medium. Supernatants of RAW 264.7 macrophage cells were collected after treatment with SMLE and exposure to LPS, then centrifuged to remove cellular debris and analysed for TNF- α , IL-1 β and IL-6 synthesis using an ELISA kit (Excell, Shanghai, China) according to the manufacturer's instructions. The absorbance was measured at 450 nm. The concentrations of TNF- α , IL-1 β and IL-6 in the experimental samples were extrapolated from a standard curve.

RNA extractions and quantitative real-time PCR (qPCR). Total cellular RNA was extracted with TRIzol reagent (TaKaRa, Seoul, Korea) according to the manufacturer's protocol. Reverse transcription was carried out at 37 °C for 1 h in a reaction mixture containing 2 μ g total RNA, 0.5 μ g oligo-dT primer, 10 mM dNTPs, 25 units RNase inhibitor and 200 units M-MLV reverse transcriptase (Promega, Madison, USA). qPCR was performed using a SYBR FAST qPCR kit (KAPA, Boston, Massachusetts, USA) with 10 ng cDNA according to the manufacturer's instructions. The cycling conditions were as follows: 95 °C for 1.5 min, and then 40 cycles of 95 °C for 3 s, 56.4 °C for 40 s and 72 °C for 10 s. Upon completion of 40 PCR amplification cycles, a melting curve analysis was performed. iNOS, TNF- α , IL-1 β and IL-6 mRNA levels were normalized to β -actin, which is a stable housekeeping gene. The primers used in this experiment were: iNOS [Fw; 5'- CTC AGC CCA ACA ATA CAA G -3'; and Rv; 5'- CTA CAG TTC CGA GCG TCA -3'], TNF- α [Fw; 5'- GCC

TAT GTC TCA GCC TCT T -3'; and Rv; 5'- GGT TGA CTT TCT CCT GGT AT -3'], IL-1 β [Fw; 5'- ATT GTG GCT GTG GAG AAG -3'; and Rv; 5'- TTG TGA GGT GCT GAT GTA -3'], IL-6 [Fw; 5'- CGA TAG TCA ATT CCA GAA ACC GC -3'; and Rv; 5'- TTG GGA GTG GTATCC TCT GTG AAG -3'] and β -actin [Fw; 5'- TGT TAC CAA CTG GGA CGA CA -3'; and Rv; 5'- AAG GAA GGC TGG AAA AGA GC -3'].

Western blot analysis. The cells were scraped off and lysed with lysis buffer (Applygen, Beijing, China). Equal amounts of protein were separated by SDS-PAGE (8% gel) and transferred to polyvinylidene difluoride (PVDF) membranes. The membranes were blocked for 1 h at room temperature with 5% nonfat dry milk, and then incubated at 4°C overnight with rabbit anti-mouse antibodies (1:1000 dilution in blocking buffer). After the membranes were washed, they were incubated with a 1:10000 dilution of horseradish peroxidase-conjugated anti-rabbit IgG secondary antibody (Jackson Immuno Research, Philadelphia, USA) for 1 h at room temperature. The blots were visualized using enhanced chemiluminescence (Applygen, Beijing, China), and data were quantified using the Gel Doc EQ System (Bio-Rad, Hercules, USA).

Intracellular ROS measurement. The production of intracellular ROS was determined using DCFH₂-DA, an oxidant-sensitive fluorescent probe, by flow cytometry as described previously (Kang *et al.*, 2002). Briefly, the cells were incubated with 10 μ M DCFH₂-DA in culture medium at 37°C for 30 min. After incubation, the cells were washed with PBS and analysed using a BD FACS Calibur flow cytometer and Cell Quest software (BD Biosciences, San Jose, CA, USA).

Intracellular glutathione (GSH) measurement. Intracellular GSH content was measured according to the reported method with slight modification (Yamada *et al.*, 2010). After incubation, the cells were deproteinized with 0.5 M HClO₄ and neutralized 2 min later with 1 M K₂CO₃. The suspensions were then centrifuged at 10000 $\times$ g for 15 min at 4°C. The supernatants (10 μ L) were transferred to 96-well black flat-bottom plates, and then 180 μ L PBS and 10 μ L o-phthalaldehyde (1 mg/mL in methanol) were added. The fluorescence was measured at an excitation wavelength of 340 nm and an emission wavelength of 405 nm using a fluorescence microplate reader (Thermo Scientific, Vantaa, Finland).

Expression of TLR4 and CD14. Briefly, RAW 264.7 cells were harvested and washed with PBS. The cells were blocked with 50 μ L of purified rat anti-mouse CD16/CD32 for 10 min to inhibit non-specific staining and then stained with optimal concentrations of TLR4/MD-2-PE or an isotype control for an additional 20 min. The antibodies to CD14-FITC and its corresponding isotype control were directly stained with cells for 20 min. Fifty thousand viable cells per treatment were analysed using a BD FACS Calibur flow cytometer and Cell Quest software.

Survival study in animal models of endotoxemia and sepsis. Endotoxemia was induced by injection of LPS (25 mg/kg, i.p.) in Balb/c mice (male, 18–20 g). In a

developing sepsis model, ICR mice (male, 20–22 g) were anesthetized with pentobarbital sodium (50 mg/kg, i.p.). Then the operation was performed according to a previously reported method (Tsoyi *et al.*, 2009). In the endotoxemia animals, 100 μ L blood samples were collected from mice by retro-orbital puncture. Serum levels of TNF- α were measured using an ELISA kit according to the manufacturer's instructions.

PMA-induced mouse ear edema. Mouse ear edema was induced with PMA according to the method reported previously by Garrido *et al.* (2004). PMA (4 μ g) dissolved in

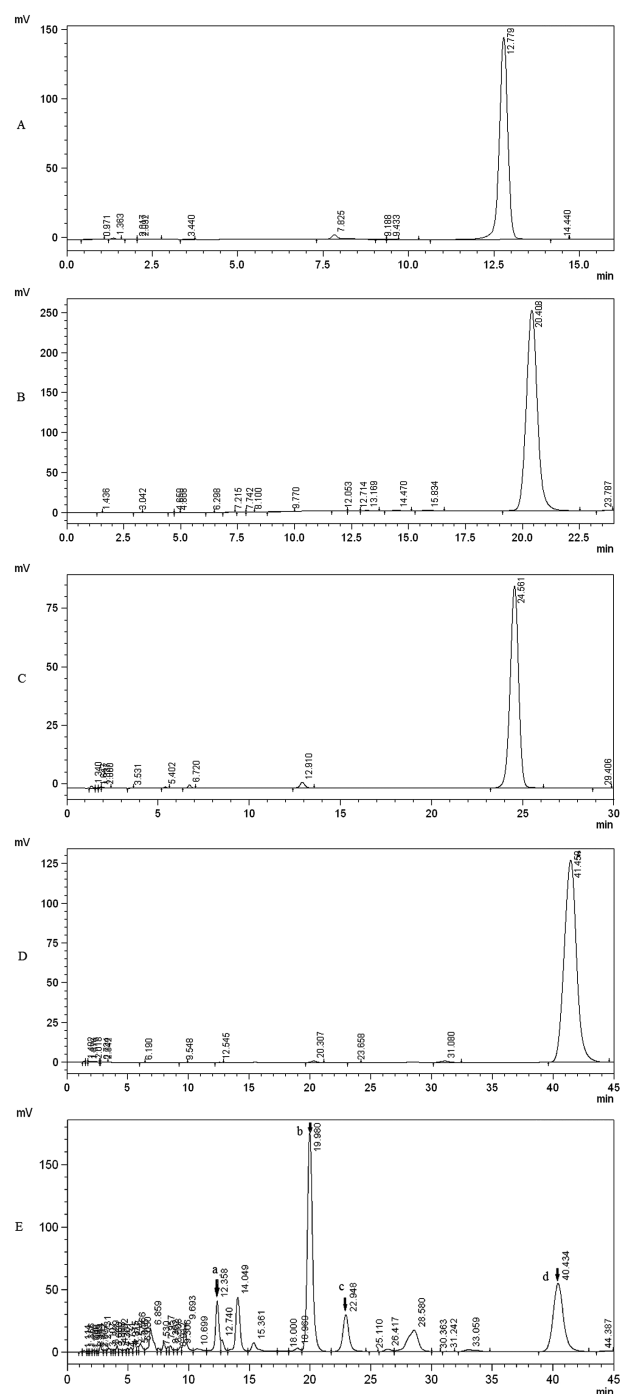


Figure 1. HPLC chromatogram of 15,16-dihydrotanshinone I (A), cryptotanshinone (B), tanshinone I (C), tanshinone IIA (D) and SMLE (E). HPLC chromatogram of 15,16-dihydrotanshinone I, cryptotanshinone, tanshinone I, tanshinone IIA in SMLE is a, b, c and d, respectively.

acetone (20 μ L) was applied to the inner and outer surfaces of the right ear of ICR mice (male, 20–22 g). SMLE was orally or topically administered to mice 1 h before PMA application. Ear biopsies were obtained with a punch (a diameter of 8 mm). Ear edema weight was calculated by subtracting the weight of the left ear from the right ear. The inhibition percentage was expressed as a reduction in weight with respect to the control group.

Statistical analysis. Results were expressed as the mean $\pm$ SD from at least three independent experiments. The statistical significance of the results between each treated group was analysed by the Student's *t*-test. Differences were considered to be significant at $p < 0.05$.

RESULTS

15,16-Dihydrotanshinone I, cryptotanshinone, tanshinone I and tanshinone IIA contents

The 15,16-dihydrotanshinone I, cryptotanshinone, tanshinone I and tanshinone IIA content of SMLE was 2.63%, 16.49%, 3.26% and 10.25%, respectively. The total content of these four compounds was 32.6% of the SMLE. The chromatogram of these four compounds in the SMLE is shown in Fig. 1.

Inhibition of NO production

The SMLE inhibited the NO production induced by LPS in RAW 264.7 cells in a dose- and time-dependent fashion without cytotoxicity (Figs 2, 3). In the sodium nitropruside (SNP) solution, the level of NO was not significantly decreased by the presence of SMLE. The incubation of RAW 264.7 macrophages, which had been pretreated with LPS for 12 h, with different concentrations of SMLE slightly affected iNOS activity in intact cells, as determined by measuring the amount of nitrite production in the medium. The mixture of the four compounds contributed less to the reduction of NO production than did the SMLE ($p < 0.01$), which contains an equal amount of these four compounds as in the mixture. The IC_{50} of the mixture is 1.47 μ g/mL and that of the SMLE is 2.29 μ g/mL, in which the total concentration of the four compounds is 0.75 μ g/mL (Fig. 3).

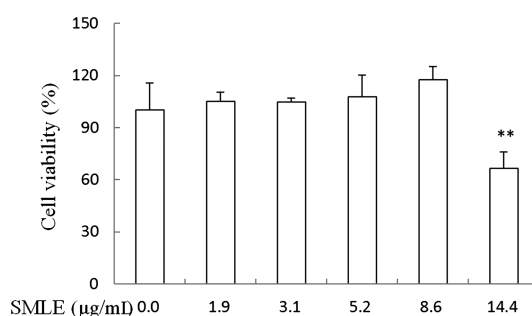


Figure 2. Effect of SMLE on the viability of RAW 264.7 macrophages. Data are expressed as mean $\pm$ SD ($n = 3$). ** $p < 0.01$ compared with the control group.

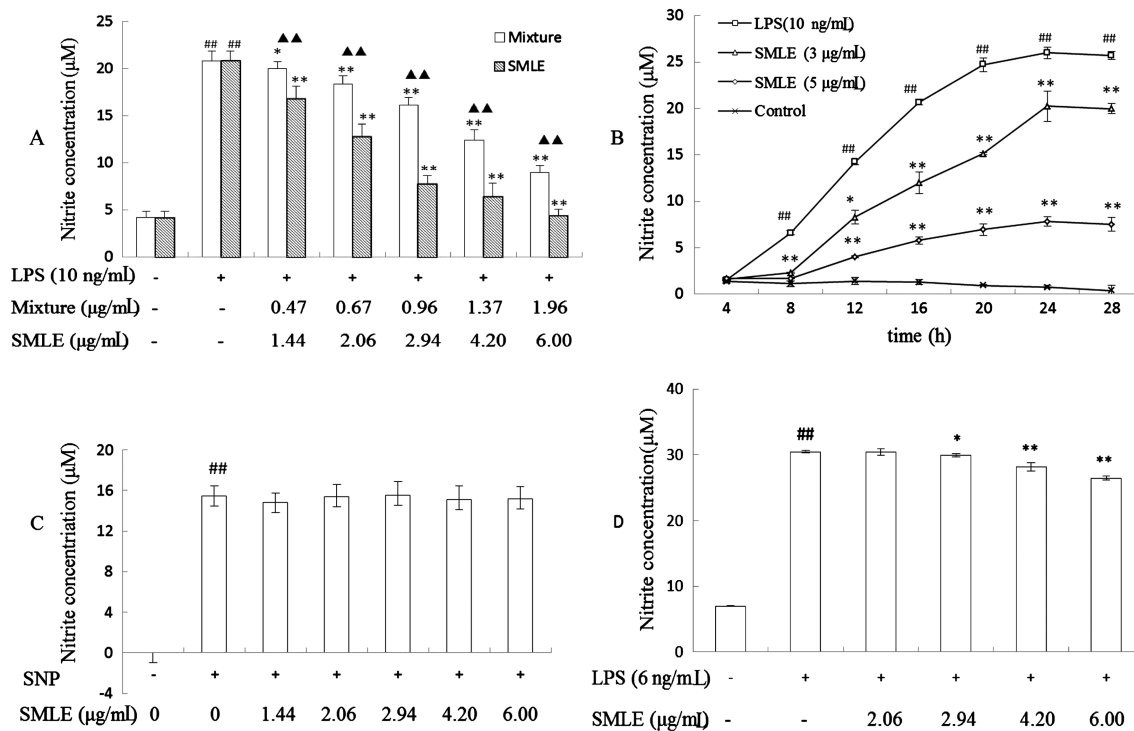


Figure 3. Effect of SMLE on NO production. (A) Cells were pretreated with the indicated concentrations of SMLE or the mixture for 2 h and then exposed to LPS (10 ng/mL) for 24 h. (B) Cells were pretreated with SMLE (3, 5 μ g/mL) for 2 h and then exposed to LPS (10 ng/mL) for 4, 8, 12, 16, 20, 24 or 28 h. (C) 60 mM SNP solution was incubated with the indicated concentrations of SMLE in the dark at 25 °C for 5 h. (D) The cells were pretreated with LPS (6 ng/mL) for 12 h. Then the cells were harvested and plated in a 48-well plate and incubated with the indicated concentrations of SMLE for another 12 h without LPS in the medium. After treatment, nitrite levels in the culture medium were measured by the Griess reaction. Data are expressed as mean $\pm$ SD ($n = 3$). ## $p < 0.01$ compared with the mock-treated group; ▲ $p < 0.01$ SMLE compared with the mixture in corresponding doses; * $p < 0.05$, ** $p < 0.01$ compared with the LPS-treated group.

Suppression of iNOS expression at both the transcriptional and translational levels in LPS-stimulated RAW 264.7 cells

To investigate the molecular mechanism by which SMLE inhibits the production of NO in response to LPS, the changes in iNOS expression following SMLE treatment were examined. As shown in Fig. 4, LPS stimulation of RAW 264.7 cells resulted in a dramatic increase in iNOS mRNA and protein expression. Pretreatment with SMLE significantly inhibited the LPS-induced iNOS mRNA and protein expression in a dose-dependent manner.

Suppression of pro-inflammatory cytokine levels in LPS-stimulated RAW 264.7 cells

The effect of SMLE on pro-inflammatory cytokine production was then examined. The RAW 264.7 cells were challenged with LPS in the presence or absence of SMLE, and the levels of TNF- α , IL-1 β and IL-6 in the medium were measured. As shown in Fig. 5, SMLE suppressed the release of TNF- α , IL-1 β and IL-6 induced by LPS into the culture supernatant in a dose-dependent manner. To determine whether the decrease in cytokine secretion was correlated with cytokine transcription

levels, we analysed the mRNA quantities of TNF- α , IL-1 β and IL-6 by qPCR. SMLE consistently down-regulated the LPS-induced transcription of TNF- α , IL-1 β and IL-6 mRNA dose-dependently.

Suppression of the I κ B- α degradation and NF- κ B nuclear translocation in LPS-stimulated RAW 264.7 cells

In order to clarify the mechanism of SMLE's inhibitory effect on the pro-inflammatory cytokines, the study investigated whether SMLE inhibits the degradation of I κ B- α and NF- κ B nuclear translocation. SMLE was shown to suppress significantly the LPS-induced degradation of I κ B- α after 30 min of LPS exposure (Fig. 6A). The nuclear translocation of p65, a component of the heterodimer of NF- κ B, was further evaluated. LPS has been shown to induce the translocation of p65 from the cytoplasm to the nucleus after 1 h of treatment, and SMLE markedly suppressed the nuclear translocation of p65 (Fig. 6B).

Effect of SMLE on intracellular ROS and GSH production in LPS-stimulated RAW 264.7 cells

To examine the effect of SMLE on intracellular ROS levels in LPS-induced cells, the production of intracellular ROS using DCFH₂-DA staining and flow cytometry was measured. LPS induced an obvious overproduction of ROS, while SMLE caused a significant decrease of ROS production in a dose-dependent manner. The intracellular GSH level was significantly decreased following LPS treatment for 24 h. SMLE significantly prevented the reduction in GSH levels, which remained close to the level of the control group at all concentrations tested (Fig. 7).

Suppression of CD14 expression in LPS-stimulated RAW 264.7 cells

The binding of LPS to CD14 and the subsequent formation of a complex with TLR4-MD-2 is the initial step in the LPS signaling pathway (Fujihara *et al.*, 2003). To examine the effect of SMLE on this step, the surface expression of CD14 and TLR4 was analysed by flow cytometry in RAW 264.7 cells stimulated with LPS. Treatment with LPS caused a marked increase in CD14 expression, and pretreatment with SMLE (6 μ g/mL) prior to LPS treatment significantly inhibited CD14 expression. Interestingly, LPS treatment reduced TLR4 expression, which is consistent with previous reports (Medvedev *et al.*, 2000; Yeo *et al.*, 2003); SMLE showed little effect on TLR4 (Fig. 8).

Protective effect of SMLE on endotoxemia or sepsis mice

In light of the capacity of SMLE to attenuate LPS-induced cytokine production and NO release *in vitro*, their efficacy in animal models of lethal endotoxemia and sepsis was explored. As shown in Fig. 9, the administration of LPS (25 mg/kg i.p.) to male Balb/c mice resulted in a survival rate of 50% after 4 days. However,

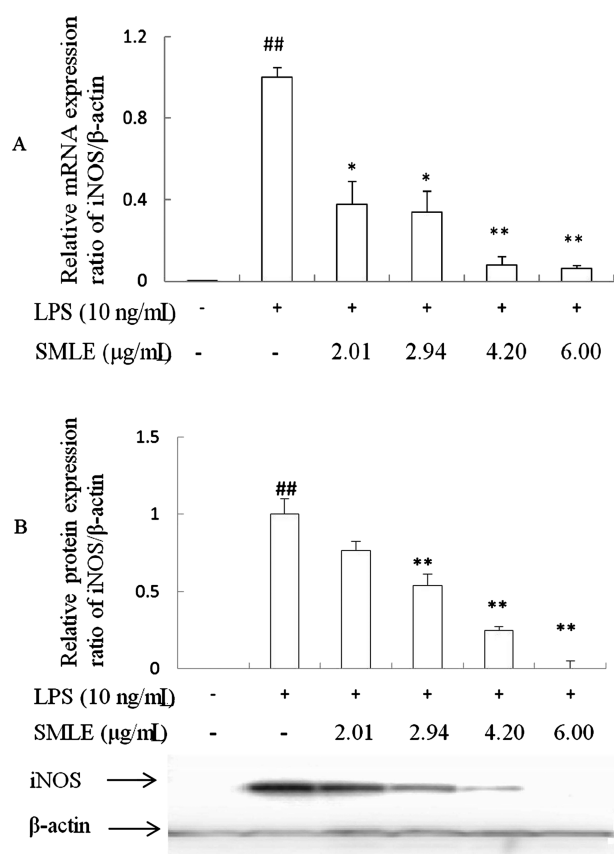


Figure 4. Effect of SMLE on iNOS mRNA and protein levels in LPS-induced RAW 264.7 cells. (A) The cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 4 h. After treatment, the total mRNA was extracted, and the mRNA levels of iNOS were determined by qPCR analysis. (B) The cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 24 h. After treatment, cellular proteins were prepared, and the protein levels of iNOS were determined by western blot analysis. Data are expressed as mean $\pm$ SD ($n = 3$). ^{##} $p < 0.01$ compared with the mock-treated group; ^{*} $p < 0.05$, ^{**} $p < 0.01$ compared with the LPS-treated group.

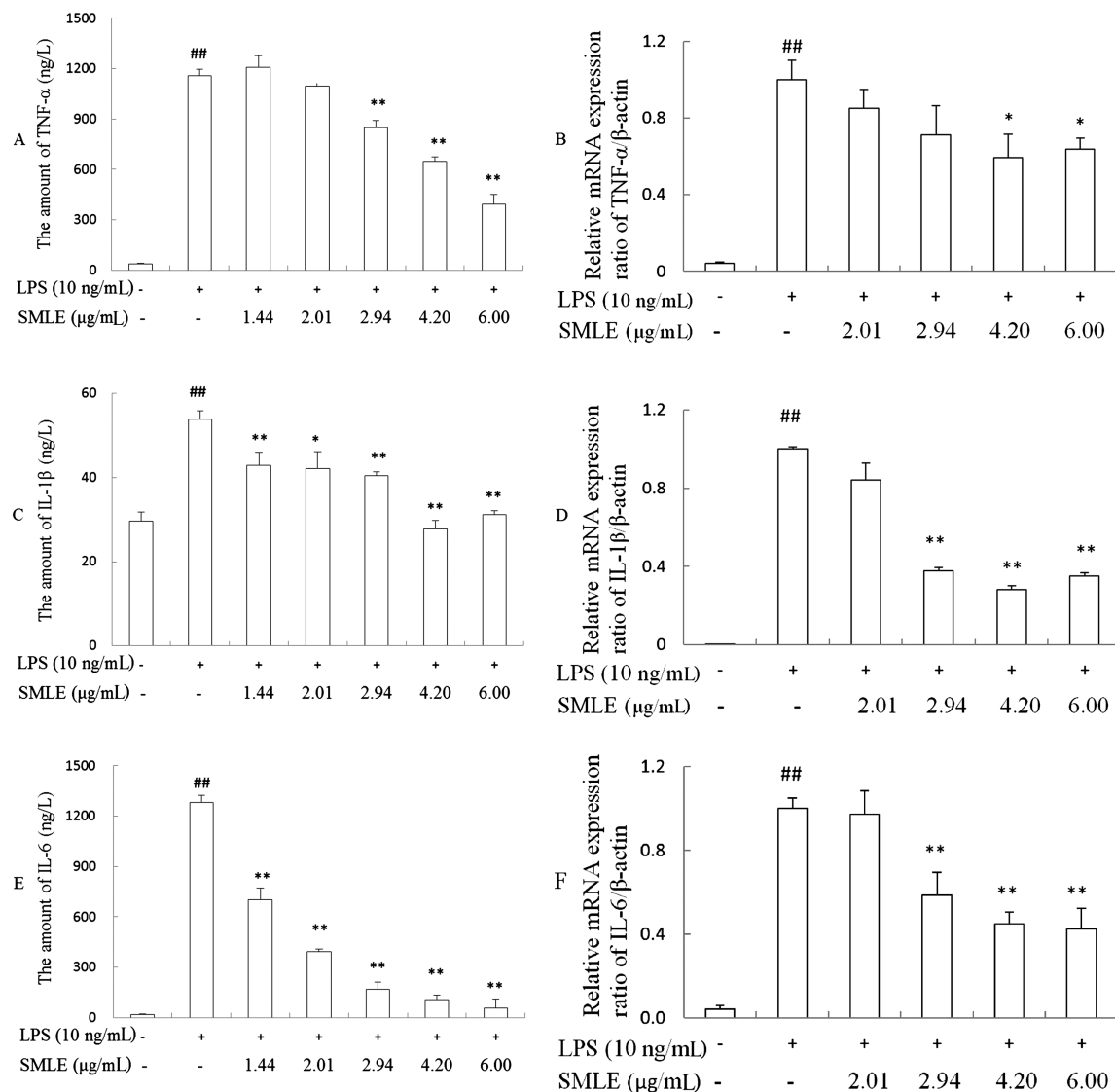


Figure 5. Effects of SMLE on TNF- α , IL-1 β and IL-6 release and their mRNA expression levels in LPS-induced RAW 264.7 cells. (A, C, E) The cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 24 h. After treatment, the TNF- α , IL-1 β and IL-6 levels in the cell culture medium were measured by ELISA. (B, D, F) The cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 4 h. After treatment, the mRNA levels of TNF- α , IL-1 β and IL-6 were determined by qPCR analysis. Data are expressed as mean $\pm$ SD ($n = 3$). ## $p < 0.01$ compared with the mock-treated group; * $p < 0.05$, ** $p < 0.01$ compared with the LPS-treated group.

treatment with SMLE (10 mg/kg i.p.) increased the survival rate to 70%, and higher levels of SMLE (40 mg/kg i.p.) nearly completely blocked LPS-induced mortality of mice (90%). The mice with cecal ligation and puncture (CLP)-induced sepsis had a survival rate of 16% after 4 days, but this rate was increased to 50% when the mice were treated with SMLE (40 mg/kg i.p.). The effect of SMLE on LPS-induced serum levels of TNF- α in Balb/c mice was also determined. Four hours after LPS injection, a significant reduction of serum TNF- α was found in SMLE-treated mice (Fig. 9).

Antiinflammatory effects of SMLE in mice with PMA-induced ear edema

Ear edema was induced by the application of PMA in mouse ears. As shown in Table 1, the application of PMA for 5 h significantly increased the ear edema. The

Table 1. Effect of SMLE on edema induced by PMA in mouse ears

Treatment	Dose	PMA-induced ear edema	
		Edema weight (mg)	Inhibition (%)
PMA (μ g/ear)	4	25.26 $\pm$ 2.71	-
SMLE (mg/ear)	0.1	17.41 $\pm$ 3.07 ^a	31.06
	0.2	16.68 $\pm$ 1.90 ^a	33.97
	0.4	18.01 $\pm$ 3.01 ^a	28.71
	0.8	15.40 $\pm$ 3.06 ^a	39.03

SMLE or mock (75% ether, 20% pyridine and 5% distilled water) was administered topically to mice 1 h before application of PMA on both surfaces of the right ears of mice. 5 h after application of PMA, ear edema and the inhibition percentage were calculated. Data are expressed as mean $\pm$ SD ($n = 8$).

^a $p < 0.01$ compared with the PMA group.

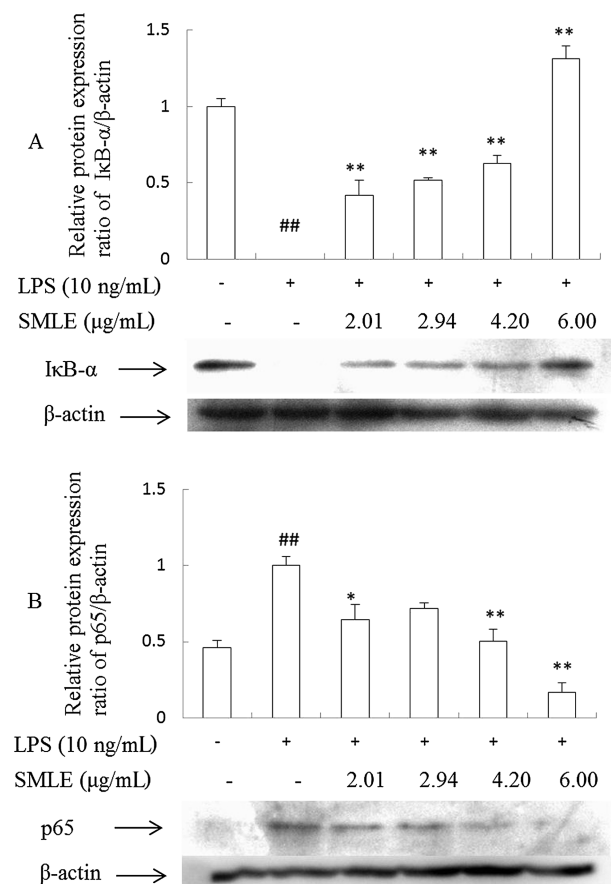


Figure 6. Effects of SMLE on $\text{IkB-}\alpha$ degradation and nuclear translocation of p65 in LPS-induced RAW 264.7 cells. (A) The cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 30 min. After treatment, cytoplasmic cellular proteins were prepared, and the protein levels of $\text{IkB-}\alpha$ were determined by western blot analysis. (B) The cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 1 h. After treatment, nuclear cellular proteins were prepared, and the protein level of p65 was determined by western blot analysis. Data are expressed as mean $\pm$ SD ($n = 3$). ^{##} $p < 0.01$ compared with the mock-treated group; ^{*} $p < 0.05$, ^{**} $p < 0.01$ compared with the LPS-treated group.

ear edema was not significantly decreased by oral administration of SMLE (100, 200 and 400 mg/kg) (data not shown). However, the topical administration of SMLE (0.1, 0.2, 0.4 and 0.8 mg/ear) significantly inhibited the PMA-induced ear edema.

DISCUSSION

The search for natural products with antiinflammatory activities has increased greatly in recent years. Herbal extracts, especially crude drugs, are known to possess a diverse range of components and secondary metabolites that have various biological activities. Although 15,16-dihydrotanshinone I, cryptotanshinone, tanshinone I and tanshinone IIA, the total content of which was 32.6% in our SMLE, were reported to have varying inhibition effects on LPS-induced NO production in RAW 264.7 cells (Jang *et al.*, 2006; Tang *et al.*, 2011; Jeon *et al.*, 2008), exact antiinflammatory studies and the underlying mechanisms of action of SMLE are poorly

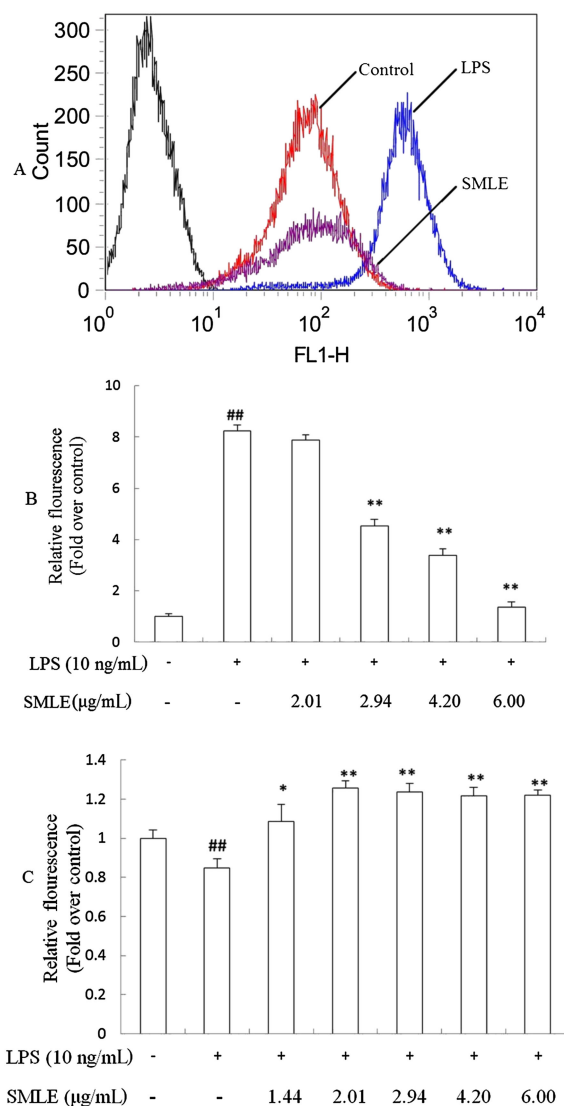


Figure 7. Effect of SMLE on intracellular ROS and GSH production in LPS-induced RAW 264.7 cells. (A) Representative flow cytometric histograms of mock-treated cells, cells exposed to LPS (10 ng/mL) for 30 min and cells pretreated with SMLE (6 $\mu\text{g/mL}$) for 2 h before exposure to LPS. (B) Fluorescent intensities of RAW 264.7 cells treated with mock, SMLE and/or LPS. (C) Cells were pretreated with the indicated concentrations of SMLE for 2 h and then exposed to LPS (10 ng/mL) for 24 h. Data are expressed as mean $\pm$ SD ($n = 3$). ^{##} $p < 0.01$ compared with the mock-treated group; ^{*} $p < 0.05$, ^{**} $p < 0.01$ compared with the LPS-treated group.

elucidated to date. By comparing the effects of the mixture of the four compounds to our SMLE (containing the four compounds equally to the mixture), it was clear that the crude extract was much more effective than the mixture in reducing NO production in LPS-induced RAW 264.7 cells. This may be due in part to the remaining 67.4% of the extract, which could also have such effects, and these two parts of the SMLE may produce an additive or even synergistic effect. Therefore, the present study was performed to investigate the anti-inflammatory activity of SMLE and to further elucidate the underlying mechanisms of its action.

Inflammation is the first response of the immune system to infection or irritation. During the inflammatory process, sizeable quantities of the pro-inflammatory mediator NO are generated by the inducible isoform of iNOS (Posadas *et al.*, 2000). Thus, agents that inhibit

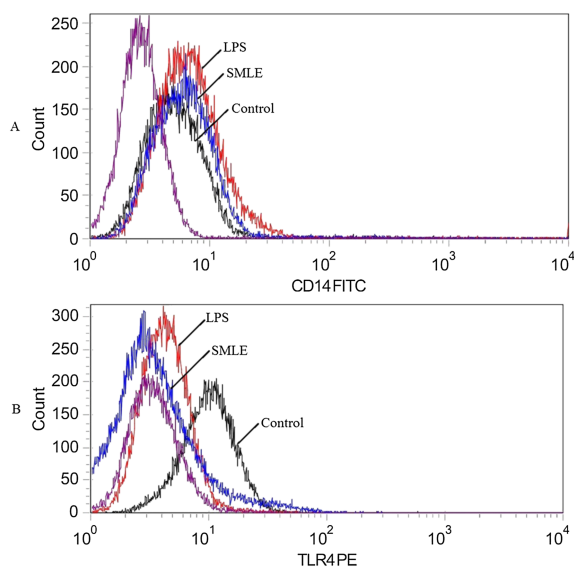


Figure 8. Effects of SMLE on CD14 and TLR4 expression in LPS-induced RAW264.7 cells. The cells were pretreated with SMLE (6 μ g/mL) for 2 h and then exposed to LPS (10 ng/mL) for 24 h. After treatment, CD14 (A) and TLR4 (B) expression in cells were measured by flow cytometry. Fluorescence histograms are shown.

the production of NO are generally considered as potential candidates for antiinflammatory drugs. SMLE inhibited LPS-induced NO production in a concentration-dependent and time-dependent manner. The down-regulation of NO by SMLE was not due to cell toxicity or a direct scavenging effect on NO but via the suppression of iNOS gene and protein expression in RAW 264.7 cells. In addition, the slight inhibition of the activity of iNOS might also contribute to the NO reduction.

Apart from NO, pro-inflammatory cytokines, such as TNF- α , IL-1 β and IL-6, play important roles in many inflammatory responses, especially septic shock (De Nardin, 2001). TNF- α , IL-1 β and IL-6 are primarily produced by macrophages, monocytes and T cells (Bondeson, 1997) and have various pro-inflammatory effects on many cell types. TNF- α plays an important role in the pathological damage of inflammation and septic shock. IL-1 β is produced early in the generation of an inflammatory response (Stylianou and Saklatvala, 1998), and IL-6 is particularly important in the production of acute-phase proteins (Diehl and Rincon, 2002). In general, they are regarded as important initiators of the inflammatory response and mediators of the development of a variety of inflammatory diseases. Thus, the modulation of cytokine production is a major mechanism in the control of inflammation. Here, SMLE resulted in a dose-dependent reduction in the LPS-induced TNF- α , IL-1 β and IL-6 release and their mRNA expression levels in RAW 264.7 cells. These findings provide evidence that SMLE acts as antiinflammatory agent influencing NO and pro-inflammatory cytokines.

NF- κ B is known to be a major transcription factor that regulates the expression of pro-inflammatory enzymes and cytokines, such as iNOS, IL-1, IL-6 and TNF- α (Chung *et al.*, 2010). NF- κ B subunits are normally sequestered in the cytosol in an inactive complex through the binding of the inhibitory factor I κ B- α in unstimulated cells. Upon stimulation of pro-inflammatory signals, including LPS, I κ B- α is phosphorylated by I κ B kinase and inactivated through ubiquitin-mediated

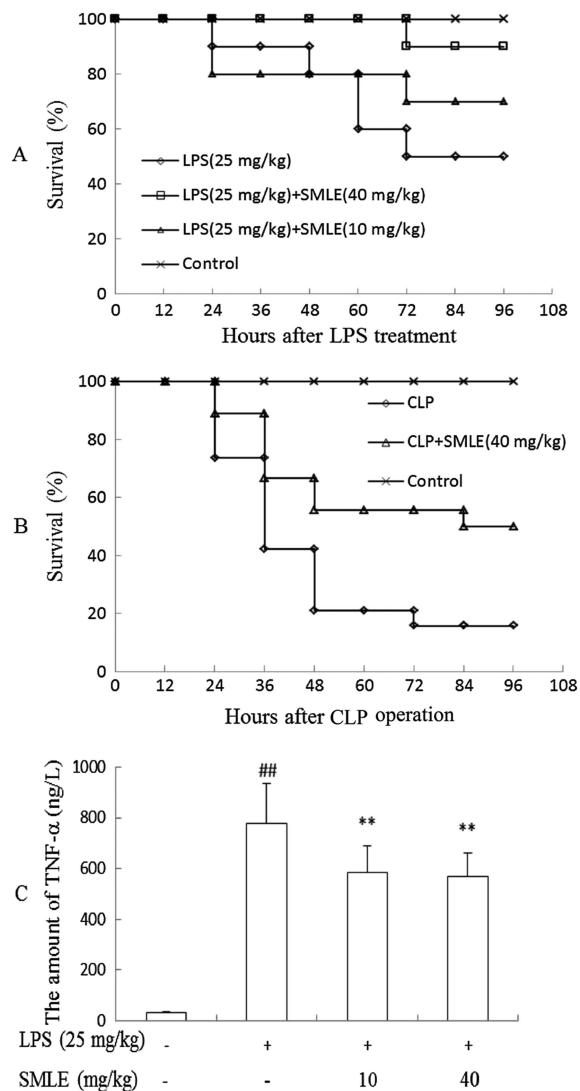


Figure 9. Effect of SMLE on the survival rate in lethal endotoxemia and sepsis in mice. (A) The mice were divided into four groups ($n = 10$), SMLE (10 mg/kg, 40 mg/kg i.p.) and mock (1% Tween 80) treatments were administered at 24 and 1 h before LPS (25 mg/kg i.p.) administration. The survival of mice was monitored every day for 4 days. (B) The mice were divided into three groups. SMLE (40 mg/kg i.p. $n = 18$), CLP (1% Tween 80 i.p. $n = 19$) and mock (1% Tween 80 i.p. $n = 20$) treatments were administered 24 and 1 h before CLP operation. At 12, 24, 48 and 72 h after the onset of sepsis, animals were treated with mock or SMLE. The survival of mice was monitored every day for 4 days. (C) 4 h after LPS treatment, the concentration of TNF- α in the serum was determined by ELISA. Data are expressed as mean $\pm$ SD ($n = 10$). $^{##}p < 0.01$ compared with the mock-treated group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with the LPS group.

degradation. The resulting free NF- κ B is translocated into the nucleus and acts as a transcription factor. The treatment with SMLE blocks the degradation of I κ B- α and nuclear translocation of p65 in LPS-induced RAW 264.7 macrophages. Therefore, these results suggest that SMLE inhibits the production of NO and pro-inflammatory cytokines through the inactivation of NF- κ B.

It is well known that changes in intracellular ROS can regulate signal transduction pathways leading to modulation of NF- κ B activity (Sanlioglu *et al.*, 2001). GSH is a crucial contributor to cellular antioxidant defense and plays a major role in protecting cells against ROS-induced oxidative injury (Schafer and Buettner, 2001; Wu *et al.*, 2004). A shortage of intracellular GSH places

cells at risk for oxidative damage (Nagai *et al.*, 2002; Satoh *et al.*, 2005; Tukov *et al.*, 2004). The data demonstrated that SMLE inhibited the production of ROS and prevented the decrease in GSH levels induced by LPS in RAW 264.7 cells. Thus, it is assumed that SMLE inhibits NF- κ B activation, at least in part, by inhibiting ROS generation and GSH depletion.

In LPS signaling, LPS first forms a complex with CD14-TLR4-MD-2. This initial complex formation is the limiting step for the intracellular signaling cascade. The binding of LPS to the CD14-TLR4-MD-2 complex triggers the activation of NF- κ B (Beutler, 2000). Therefore, the study investigated whether SMLE could inhibit the formation of this complex. Flow cytometric analysis revealed that SMLE inhibited the expression of binding receptors of CD14 in LPS-treated RAW 264.7 cells. Thus, the inhibition of the binding of LPS to cells by SMLE may be another pathway leading to the inhibition of NF- κ B activation.

In *in vivo* studies, SMLE showed a protective effect on mice with LPS-induced endotoxemia and CLP-induced sepsis. Further studies also demonstrated that the increase in serum levels of TNF- α caused by LPS was down-regulated by SMLE intraperitoneal treatment. Topical application of PMA induced an inflammatory response. In the indicated doses, topical administration of SMLE showed an inhibitory effect on PMA-induced edemas, while oral administration did not, suggesting that SMLE may have a low oral bioavailability. As is known, cryptotanshinone is a substrate for P-glycoprotein (Pgp), which can pump its substrate

into the luminal side of small intestine (Zhang *et al.*, 2006). As cryptotanshinone comprises 16.49% of our SMLE, it is worth investigating whether other effective compounds are in a similar situation.

In summary, it was demonstrated that SMLE inhibits LPS-induced gene and protein expression of iNOS, TNF- α , IL-1 β and IL-6 in macrophages, and these effects are mediated by blocking NF- κ B activation. The inhibitory effect of SMLE on NF- κ B activation may be due to the reduction of intracellular ROS and inhibition of surface CD14 expression. Additionally, SMLE showed a protective effect on endotoxemia and sepsis animals and an inhibition effect on ear edema models. The route of SMLE administration is closely related to the drug efficacy, which is a finding of critical importance for clinical applications.

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Conflict of Interest

The authors have declared that there is no conflict of interest.

REFERENCES

- Adams DO, Hamilton TA. 1984. The cell biology of macrophage activation. *Annu Rev Immunol* **2**: 283–318.
- Allen RG, Tresini M. 2000. Oxidative stress and gene regulation. *Free Radic Biol Med* **28**: 463–499.
- Beutler B. 2000. Tlr4: central component of the sole mammalian LPS sensor. *Curr Opin Immunol* **12**: 20–26.
- Bondeson J. 1997. The mechanisms of action of disease-modifying antirheumatic drugs: a review with emphasis on macrophage signal transduction and the induction of proinflammatory cytokines. *Gen Pharmacol* **29**: 127–150.
- Chung HJ, Lee HS, Shin JS, *et al.* 2010. Modulation of acute and chronic inflammatory processes by a traditional medicine preparation GCSB-5 both *in vitro* and *in vivo* animal models. *J Ethnopharmacol* **130**: 450–459.
- De Nardin E. 2001. The role of inflammatory and immunological mediators in periodontitis and cardiovascular disease. *Ann Periodontol* **6**: 30–40.
- Diehl S, Rincón M. 2002. The two faces of IL-6 on Th1/Th2 differentiation. *Mol Immunol* **39**: 531–536.
- Fujihara M, Muroi M, Tanamoto K, Suzuki T, Azuma H, Ikeda H. 2003. Molecular mechanisms of macrophage activation and deactivation by lipopolysaccharide: roles of the receptor complex. *Pharmacol Ther* **100**: 171–194.
- Garrido G, González D, Lemus Y, *et al.* 2004. *In vivo* and *in vitro* anti-inflammatory activity of *Mangifera indica* L. extract (VIMANG). *Pharmacol Res* **50**: 143–149.
- Gius D, Botero A, Shah S, Curry HA. 1999. Intracellular oxidation/reduction status in the regulation of transcription factors NF- κ B and AP-1. *Toxicol Lett* **106**: 93–106.
- Jang SI, Kim HJ, Kim YJ, Jeong SI, You YO. 2006. Tanshinone IIA inhibits LPS-induced NF- κ B activation in RAW 264.7 cells: Possible involvement of the NIK–IKK, ERK1/2, p38 and JNK pathways. *Eur J Pharmacol* **542**: 1–7.
- Jeon SJ, Son KH, Kim YS, Choi YH, Kim HP. 2008. Inhibition of prostaglandin and nitric oxide production in lipopolysaccharide-treated RAW 264.7 cells by tanshinones from the roots of *Salvia miltiorrhiza* Bunge. *Arch Pharmacol Res* **31**: 758–763.
- Kang JS, Jeon YJ, Kim HM, Han SH, Yang KH. 2002. Inhibition of inducible nitric-oxide synthase expression by silymarin in lipopolysaccharide-stimulated macrophages. *J Pharmacol Exp Ther* **302**: 138–144.
- Medvedev AE, Kopydlowski KM, Vogel SN. 2000. Inhibition of lipopolysaccharide-induced signal transduction in endotoxin-tolerized mouse macrophages: dysregulation of cytokine, chemokine and toll-like receptor 2 and 4 gene expression. *J Immunol* **164**: 5564–5574.
- Nagai H, Matsumaru K, Feng G, Kaplowitz N. 2002. Reduced glutathione depletion causes necrosis and sensitization to tumor necrosis factor- α -induced apoptosis in cultured mouse hepatocytes. *Hepatology* **36**: 55–64.
- Nam NH. 2006. Naturally occurring NF- κ B inhibitors. *Mini Rev Med Chem* **6**: 945–951.
- Park EJ, Zhao YZ, Kim YC, Sohn DH. 2009. Preventive effects of a purified extract isolated from *Salvia miltiorrhiza* enriched with tanshinone I, tanshinone IIA and cryptotanshinone on hepatocyte injury *in vitro* and *in vivo*. *Food Chem Toxicol* **47**: 2742–2748.
- Posadas I, Terencio MC, Guillén I, *et al.* 2000. Co-regulation between cyclo-oxygenase-2 and inducible nitric oxide synthase expression in the time-course of murine inflammation. *Naunyn Schmiedeberg's Arch Pharmacol* **361**: 98–106.
- Sanlioglu S, Williams CM, Samavati L, *et al.* 2001. Lipopolysaccharide induces Rac1-dependent reactive oxygen species formation and coordinates tumor necrosis factor- α secretion through IKK regulation of NF- κ B. *J Biol Chem* **276**: 30188–30198.
- Satoh E, Okada M, Takadera T, Ohyashiki T. 2005. Glutathione depletion promotes aluminum-mediated cell death of PC12 cells. *Biol Pharm Bull* **28**: 941–946.
- Schafer FQ, Buettner GR. 2001. Redox environment of the cell as viewed through the redox state of the glutathione disulfide/glutathione couple. *Free Radic Biol Med* **30**: 1191–1212.
- Serra S, Vacca G, Tumat S, *et al.* 2003. Anti-relapse properties of IDN 5082, a standardized extract of *Salvia miltiorrhiza*, in alcohol-preferring rats. *J Ethnopharmacol* **88**: 249–252.

- Shi Z, He J, Yao T, Chang W, Zhao M. 2005. Simultaneous determination of cryptotanshinone, tanshinone I and tanshinone IIA in traditional Chinese medicinal preparations containing *Radix salvia miltiorrhiza* by HPLC. *J Pharm Biomed Anal* **37**: 481–486.
- Stylianou E, Saklatvala J. 1998. Interleukin-1. *Int J Biochem Cell Biol* **30**: 1075–1079.
- Tang S, Shen XY, Huang HQ, *et al.* 2011. Cryptotanshinone suppressed inflammatory cytokines secretion in RAW 264.7 macrophages through inhibition of the NF- κ B and MAPK signaling pathways. *Inflammation* **34**: 111–118.
- Tsoyi K, Lee TY, Lee YS, *et al.* 2009. Heme-oxygenase-1 induction and carbon monoxide-releasing molecule inhibit lipopolysaccharide (LPS)-induced high-mobility group box 1 release *in vitro* and improve survival of mice in LPS- and cecal ligation and puncture-induced sepsis model *in vivo*. *Mol Pharmacol* **76**: 173–182.
- Tukov FF, Rimoldi JM, Matthews JC. 2004. Characterization of the role of glutathione in repin-induced mitochondrial dysfunction, oxidative stress and dopaminergic neurotoxicity in rat pheochromocytoma (PC12) cells. *Neurotoxicology* **25**: 989–999.
- Wang GJ, Chen SM, Chen WC, Chang YM, Lee TH. 2007. Selective inducible nitric oxide synthase suppression by new bracteanolides from *Murdannia bracteata*. *J Ethnopharmacol* **112**: 221–227.
- Wu G, Fang YZ, Yang S, Lupton JR, Turner ND. 2004. Glutathione metabolism and its implications for health. *J Nutr* **134**: 489–492.
- Yamada T, Egashira N, Imuta M, *et al.* 2010. Role of oxidative stress in vinorelbine-induced vascular endothelial cell injury. *Free Radic Biol Med* **48**: 120–127.
- Yang W, Ju JH, Jeon MJ, Han X, Shin I. 2010. Danshen (*Salvia miltiorrhiza*) extract inhibits proliferation of breast cancer cells via modulation of Akt activity and p27 level. *Phytother Res* **24**: 198–204.
- Yeo SJ, Yoon JG, Hong SC, Yi AK. 2003. CpG DNA induces self and cross-hyporesponsiveness of RAW 264.7 cells in response to CpG DNA and lipopolysaccharide: alterations in IL-1 receptor-associated kinase expression. *J Immunol* **170**: 1052–1061.
- Zhang J, Huang M, Guan S, *et al.* 2006. A mechanistic study of the intestinal absorption of cryptotanshinone, the major active constituent of *Salvia miltiorrhiza*. *J Pharmacol Exp Ther* **317**: 1285–1294.
- Zhang Q, Chang Z, Yang J, Wang Q. 2008. Anti atherogenic property of triterpenoids-enriched extract from the aerial parts of *Salvia miltiorrhiza*. *Phytother Res* **22**: 1040–1045.
- Zhonghua Bencao Editors. 1998. *Zhonghua Bencao (Chinese Herbal Medicine)*. Shanghai Science and Technology Publishing House: Shanghai, 1663–1664.